BORONG XU

Computer Engineering Student

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ABOUT ME

- Undergraduate Computer Engineering student at HKUST, passionate about robotics and embodied intelligence. Skilled in C/C++, Python, and ROS with hands-on project experience. Eager to contribute to intelligent robot systems in real-world applications.

EDUCATION

The Hong Kong University of Science and Technology (HKUST) — *Hong Kong SAR*

Bachelor of Engineering in Computer Engineering — *Expected Graduation: June 2026*

- GPA 3.9/4.3 (Top 3% among 150 students)
- Major GPA 4.1/4.3

High School Affiliated to Renmin University of China (RDFZ) — *Beijing, China*

High School Diploma(GaoKao) — *Graduated June 2022*

- Activities:
  - FIRST Robotics Club, Chief Engineer (season 2021)
  - FIRST Robotics Club, Alumni Mentor (season 2022-2024)

PROJECT EXPERIENCE

LLM Agent Development — *Supervised by Prof. Shenghui Song*

Undergraduate Research Opportunity (UROP) — *Jun 2024 - May 2026*

- Designed and implemented a Large Language Model (LLM) agent capable of playing board games. Conducted comprehensive evaluations of agent strategies and performance.

ELEC5660 Introduction to Aerial Robotics — *Supervised by Prof. Shaojie Shen*

Course Project — *Feb 2026 - May 2026*

- Developed aerial robot algorithms from scratch, including visual-inertial odometry, Extended Kalman Filter (EKF) state estimation, and autonomous path planning within a simulation environment, enabling robust UAV localization, mapping, and navigation.

Quadruped Robot Control — *Supervised by Prof. Maurice Pagnucco and Prof. Yang Song*

Research Practicum — *Feb 2025 - Aug 2025*

- Built and programmed a Unitree GO2 quadruped robot to perform complex tasks, integrating depth camera for accurate object localization (localization error < 10cm) and LiDAR for SLAM.

Underwater Localization System — *Supervised by Prof. Huan Yin and Prof. Fumin Zhang*

Final Year Project — *Jun 2025 - May 2026*

- Developed a Multi-Camera Vision-Based Localization System for Tracking Multiple Autonomous Underwater Vehicles (AUVs). Reduced localization error in complex underwater lighting scenarios, improving positioning accuracy to within 10cm in a 1m×1m range.

HONORS

2022-2026	**Dean's List,** 5 times	*HKUST*
2023-2026	**Continuing Undergraduate Students Scholarship,** 10,000 HKD per year	*HKUST*
2023	**Bronze Medal,** The Innovation Competition: Our Livable City, We Engineer	*HKIE*
2022	**Entry Scholarship,** 70,000 HKD	*HKUST*

SKILLS

- **C/C++**: Eigen, PCL, STL
- **Python**: PyTorch, OpenCV, Numpy
- **ROS**: Navigation2, Gazebo, Slam Toolbox, TF

LANGUAGES

English • Fluent (IELTS 7.5)
Mandarin • Mother tongue

NATIONALITY

Chinese